

SOLUTIONS

Below, you can find solutions for each task. Remember though, that you can follow your own strategy, which may be different from the one explained in the following lab.

Task 1. Have malicious WMI commands been executed on the machine?

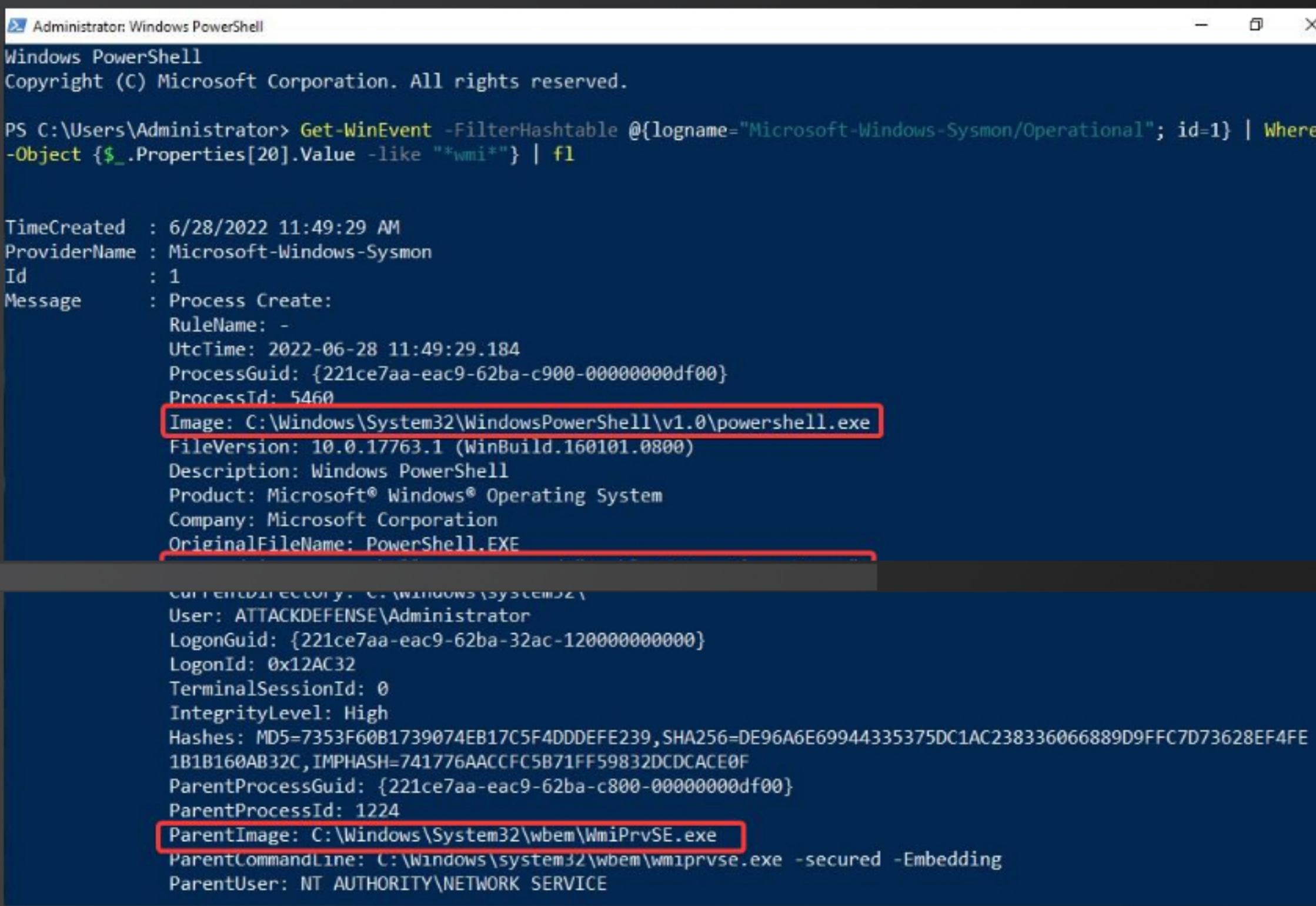
After reading technical reports (a.k.a tactical threat intelligence) on WMIImplant, we can identify the below artifacts/traces.

- wmiprvse.exe spawning powershell.exe and probably executing obfuscated commands
- Environment Variable name on EIDs 12 and 13; Pattern: `\\Environment\\[a-zA-Z0-9]{5}`
- by PowerShell.exe

Now for our hunt, we can take an approach where we either group the indicators above, or we do a search based only on one of them. Since we are dealing with a single source of logs, we can take the second approach (this can be tedious though when dealing with big environments and multi terabytes of logs). Let's begin by the first indicator and check if any processes have been spawned by "wmiprvse.exe". To parse for that, run the following command (inside a PowerShell as an administrator terminal):

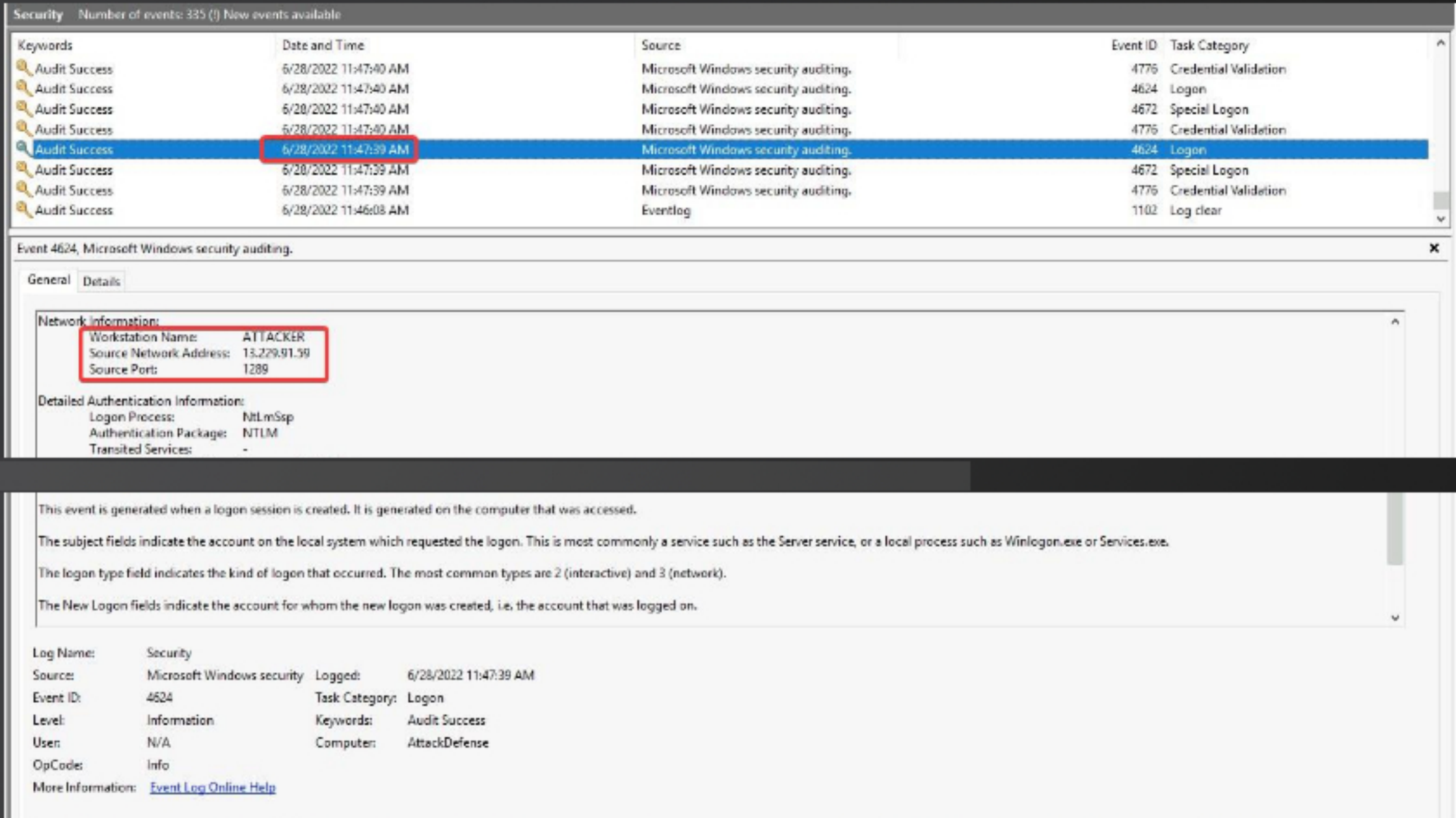
```
Get-WinEvent -FilterHashtable @{{Logname="Microsoft-Windows-Sysmon/Operational"; id=1} | Where
```

The output of the command contains only 1 result, which looks as follows:

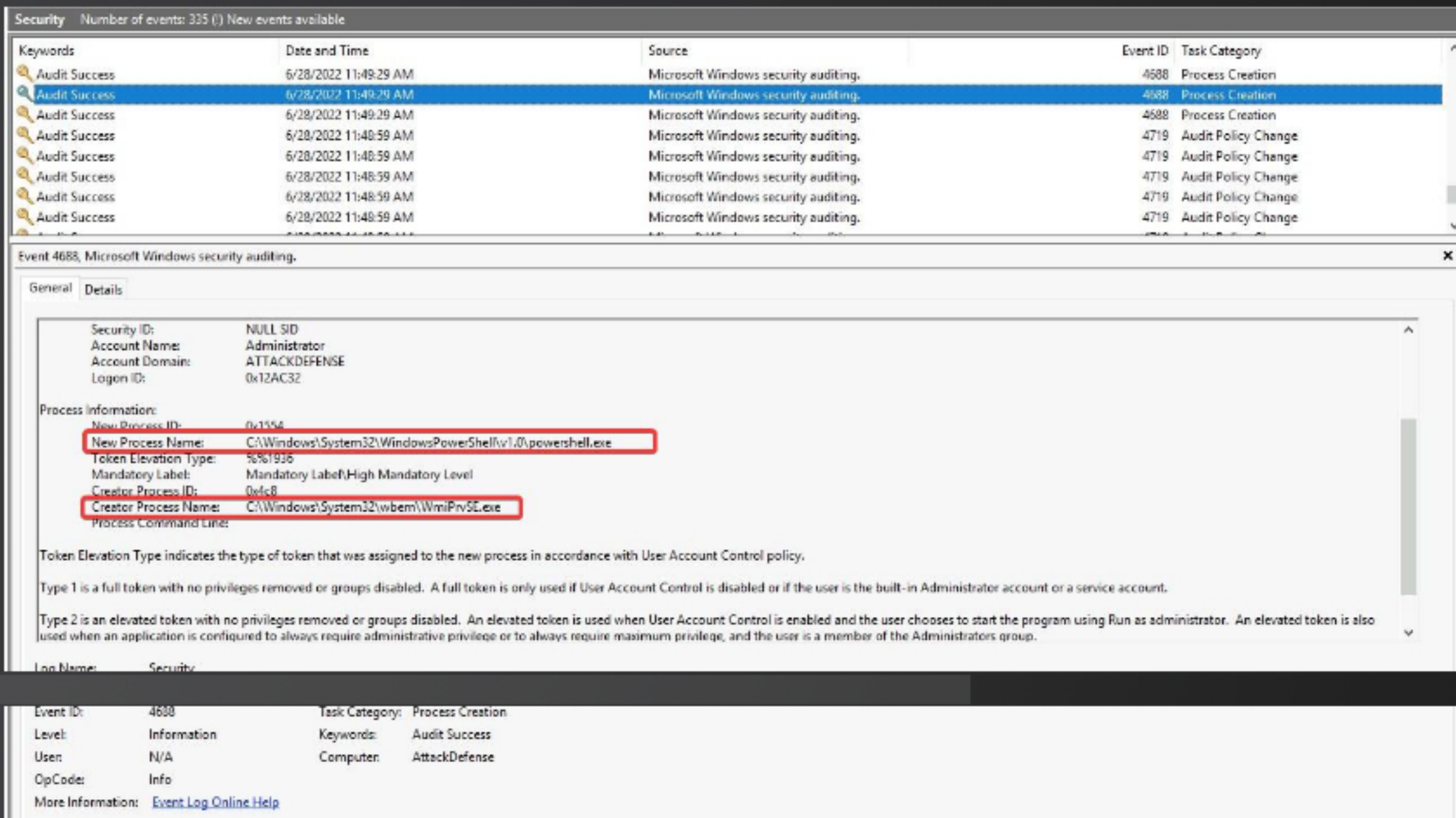


In fact, it appears that WmiPrvSE.exe has been executed on the machine, which spawns PowerShell with a command line parameter that aims to enable PowerShell Remoting.

We can "chain" that with Windows Security Logs, where we can see a login of the "Administrator" account from a remote machine with IP "13.229.91.59":



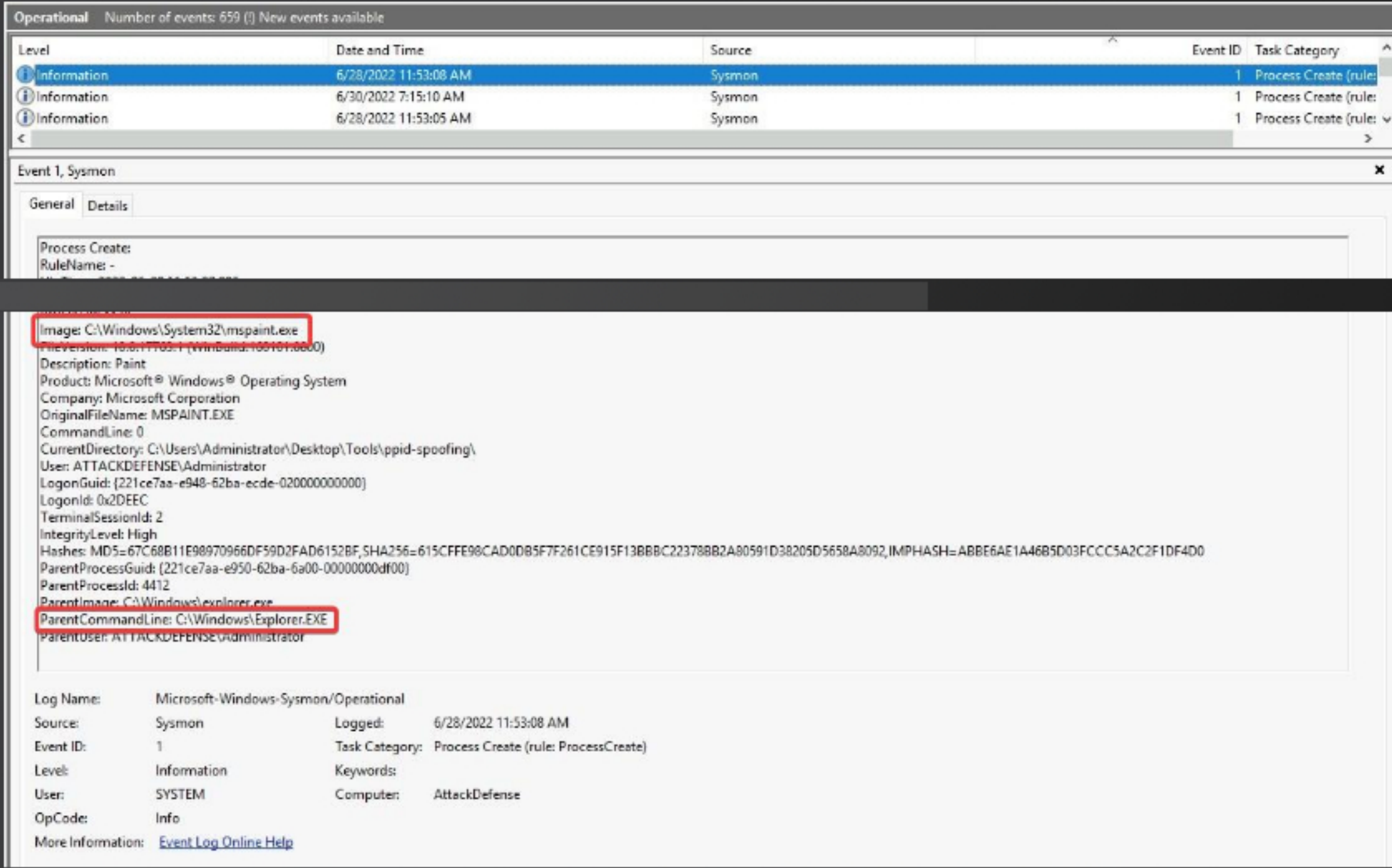
After which, the PowerShell Process is spawned:



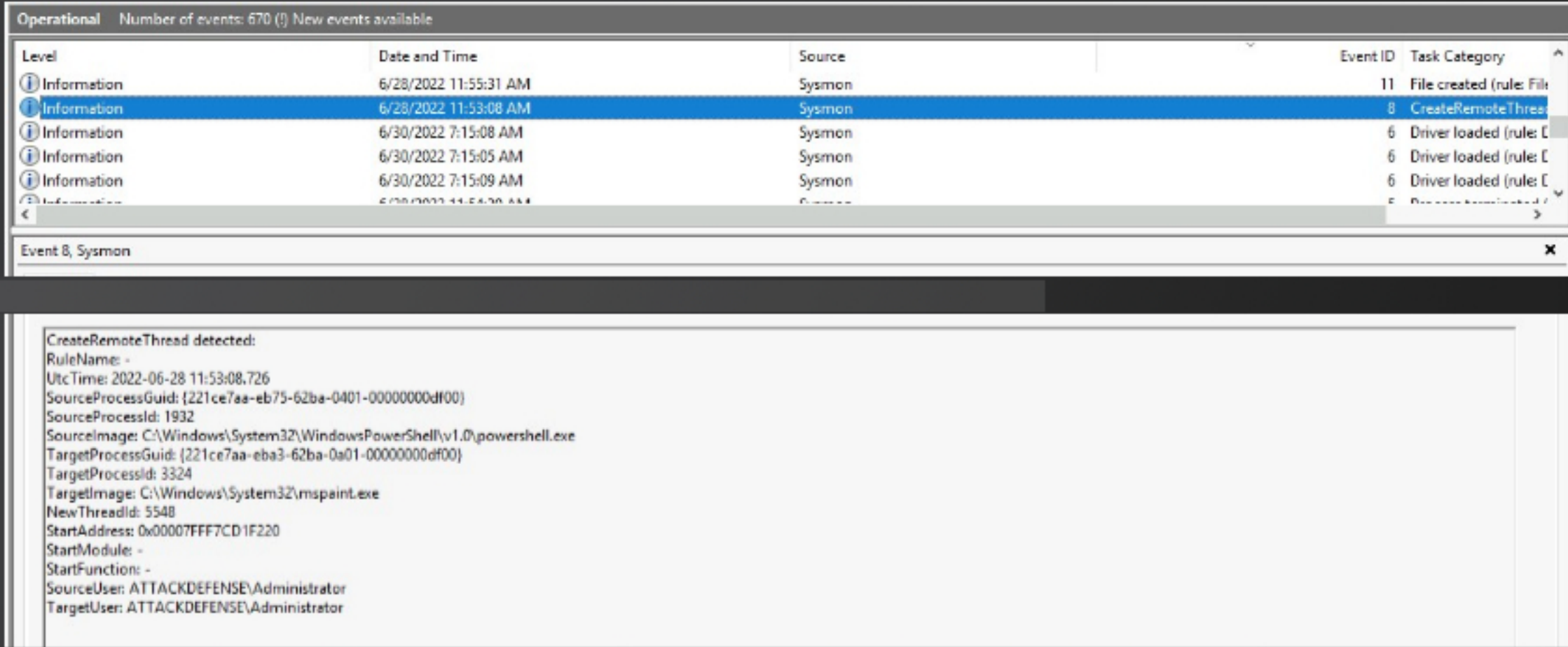
Note - If you look for other artifacts that we listed earlier, you may not find any of them because the red team did not utilize all of WMIImplant's capabilities - perhaps all that was required was a single action enabling PSRemoting. We would have missed this if we had searched only for occurrence of, for example, 2 out of the 3 detection rules. Have this in mind during your hunts. Don't always search for multiple indicators at once!

Task 2. Has Parent Process Spoofing occurred on the machine?

By consulting with Sysmon's Process Creation logs or with the Security Process Creation logs, we can identify the following.



explorer.exe seems to be starting mspaint.exe. This looks quite natural, but an interesting event occurs immediately right after, where PowerShell creates a remote thread to mspaint.exe (which, to some extent, indicates that there is some kind of interaction between the 2 processes):



KEY TAKEAWAYS

- Robust detection rules and non-default audit policies are required to perform high quality hunts
- Adversaries often choose to hide in plain sight
- Defender tools can be abused or disabled by attackers